Radios

A radio is a device that intercepts radio waves and converts them into a useful form. Broadcast radios receive audio programs transmitted by radio stations and play them through speakers.

How a radio works

A radio receives radio waves through an antenna, which converts them into small alternating currents. These currents are applied to a receiver, which filters out unwanted frequencies and amplifies the signals. The signals are then demodulated: the useful information-carrying signal is extracted from the carrier wave with which it was combined for transmission (see pp.180–181). Finally, the original audio program is played through speakers. Very simple radio receivers (tuned radio frequency receivers) carry out just these steps, but most radios perform additional processing.

STATIC IS CAUSED BY AMPLIFYING **RANDOM ELECTRICAL SIGNALS** BETWEEN BROADCAST FREQUENCIES

